

Objective / Aim

To design and implement a **student performance analysis system** in Java using **Object-Oriented Programming principles** to calculate marks, percentage, grade, highest and lowest marks, and overall academic performance.

Problem Statement

Develop a Java application for a college to analyze the academic performance of a student. The system should accept the student's **details and marks**. It should perform the following operations:

- Store subject marks using an array.
- Calculate total and average marks.
- Find the highest and lowest marks.
- Calculate percentage.
- Assign a grade based on percentage.
- Determine whether the student has passed or failed.
- Display the student's name in a formatted manner using String operations.
- Round the percentage to two decimal places using the Math package.
- Display a simple performance remark based on the student's percentage.
- The program should use a **class** to represent the student and demonstrate basic object-oriented programming.

Algorithm / Logic Description Create a class named Student. Declare the following data members:

- Roll number
- Student name
- Array of marks

Create a constructor to initialize the student details. Create methods to:

- Calculate total marks.
- Calculate average marks.
- Find highest marks.
- Find lowest marks.
- Calculate percentage.
- Determine grade.

- Display student details.

Store marks of five subjects in an integer array. Use a loop to calculate the total marks. Use `Math.max()` and `Math.min()` or suitable logic to determine the highest and lowest marks. Calculate the percentage from the total marks. Use `Math.round()` to display the percentage up to two decimal places. Use String operations such as `trim()`, `toUpperCase()`, and `length()` to process the student's name. Assign grades using the following criteria:

90–100 A+

80–89 A

70–79 B

60–69 C

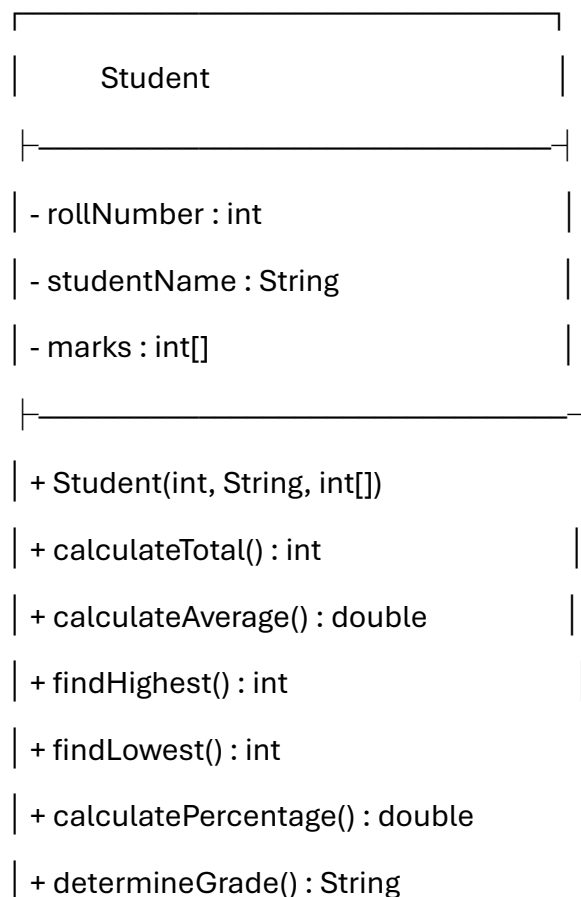
50–59 D

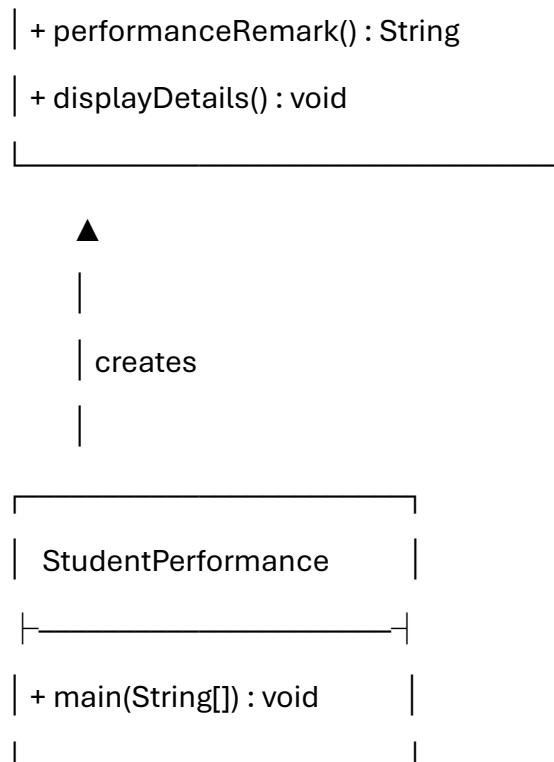
Below 50 F (Fail) Consider the student passed only if the percentage is at least 50%. Display a performance remark based on the grade. Create a Student object in the `main()` method. Display the complete performance report.

Real-time application

College student performance analysis

UML Diagram:





Code:

```
package Observations;
```

```
import java.util.Scanner;
```

```
class Student {
```

```
    int rollNumber;
```

```
    String studentName;
```

```
    int[] marks = new int[5];
```

```
    // Constructor
```

```
    Student(int rollNumber, String studentName, int[] marks) {
```

```
        this.rollNumber = rollNumber;
```

```
        this.studentName = studentName.trim();
```

```
        this.marks = marks;
```

```
    }
```

```
// Calculate total marks

int calculateTotal() {
    int total = 0;

    for (int i = 0; i < marks.length; i++) {
        total += marks[i];
    }

    return total;
}

// Calculate average marks

double calculateAverage() {
    return calculateTotal() / 5.0;
}

// Find highest marks

int findHighest() {
    int highest = marks[0];

    for (int i = 1; i < marks.length; i++) {
        highest = Math.max(highest, marks[i]);
    }

    return highest;
}

// Find lowest marks
```

```
int findLowest() {  
    int lowest = marks[0];  
  
    for (int i = 1; i < marks.length; i++) {  
        lowest = Math.min(lowest, marks[i]);  
    }  
  
    return lowest;  
}  
  
// Calculate percentage  
double calculatePercentage() {  
    return (calculateTotal() / 500.0) * 100;  
}  
  
// Determine grade  
String determineGrade() {  
    double percentage = calculatePercentage();  
  
    if (percentage >= 90)  
        return "A+";  
    else if (percentage >= 80)  
        return "A";  
    else if (percentage >= 70)  
        return "B";  
    else if (percentage >= 60)  
        return "C";  
    else if (percentage >= 50)
```

```
        return "D";
    else
        return "F";
}
```

```
// Performance remark
```

```
String performanceRemark() {
```

```
    String grade = determineGrade();
```

```
    switch (grade) {
```

```
        case "A+":
```

```
            return "Excellent Performance";
```

```
        case "A":
```

```
            return "Very Good Performance";
```

```
        case "B":
```

```
            return "Good Performance";
```

```
        case "C":
```

```
            return "Satisfactory Performance";
```

```
        case "D":
```

```
            return "Needs Improvement";
```

```
        default:
```

```
            return "Fail - Work Hard";
```

```
    }
```

```
}
```

```
// Display student details
```

```
void displayDetails() {
```

```
    double percentage = calculatePercentage();
```

```

// Math.round() to two decimal places

double roundedPercentage = Math.round(percentage * 100.0) / 100.0;

System.out.println("\n STUDENT PERFORMANCE REPORT \n");

System.out.println("Roll Number    : " + rollNumber);

System.out.println("Student Name    : " + studentName.toUpperCase());

System.out.println("Name Length    : " + studentName.length());

System.out.println("\nSubject Marks:");

for (int i = 0; i < marks.length; i++) {

    System.out.println("Subject " + (i + 1) + "      : " + marks[i]);

}

System.out.println("\nTotal Marks    : " + calculateTotal() + " / 500");

System.out.println("Average Marks   : " + calculateAverage());

System.out.println("Highest Marks   : " + findHighest());

System.out.println("Lowest Marks    : " + findLowest());

System.out.println("Percentage      : " + roundedPercentage + "%");

System.out.println("Grade           : " + determineGrade());

if (percentage >= 50)

    System.out.println("Result          : PASS");

else

    System.out.println("Result          : FAIL");

System.out.println("Remark          : " + performanceRemark());

}

}

public class StudentPerformance {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

```

```
System.out.print("Enter Roll Number: ");
```

```
int rollNumber = sc.nextInt();
```

```
sc.nextLine();
```

```
System.out.print("Enter Student Name: ");
```

```
String studentName = sc.nextLine();
```

```
int[] marks = new int[5];
```

```
System.out.println("Enter marks for 5 subjects:");
```

```
for (int i = 0; i < 5; i++) {
```

```
    System.out.print("Subject " + (i + 1) + ": ");
```

```
    marks[i] = sc.nextInt();
```

```
}
```

```
// Create Student object
```

```
Student student = new Student(rollNumber, studentName, marks);
```

```
// Display complete performance report
```

```
student.displayDetails();
```

```
sc.close();
```

```
}
```

```
}
```

Output:

```

es -cp C:\Users\Sahithi\pyla\AppData\Roaming\Code\User\wo
bservations.StudentPerformance'
Enter Roll Number: 123
Enter Student Name: sakshi
Enter marks for 5 subjects:
Subject 1: 96
Subject 2: 97
Subject 3: 93
Subject 4: 94
Subject 5: 90

===== STUDENT PERFORMANCE REPORT =====
Roll Number      : 123
Student Name     : SAKSHI
Name Length      : 6

Subject Marks:
Subject 1        : 96
Subject 2        : 97
Subject 3        : 93
Subject 4        : 94
Subject 5        : 90

Total Marks      : 470 / 500
Average Marks    : 94.0
Highest Marks    : 97
Lowest Marks     : 90
Percentage       : 94.0%
Grade            : A+
Result           : PASS
Remark           : Excellent Performance
PS C:\Users\Sahithi\pyla\Desktop\Sahithi java>

```

Result:

The **Student Performance Analysis System** was successfully implemented in Java using **OOP concepts, arrays, and built-in packages**. The program successfully calculates and displays the student's academic performance, including total marks, average, highest and lowest marks, percentage, grade, result, and performance remark.